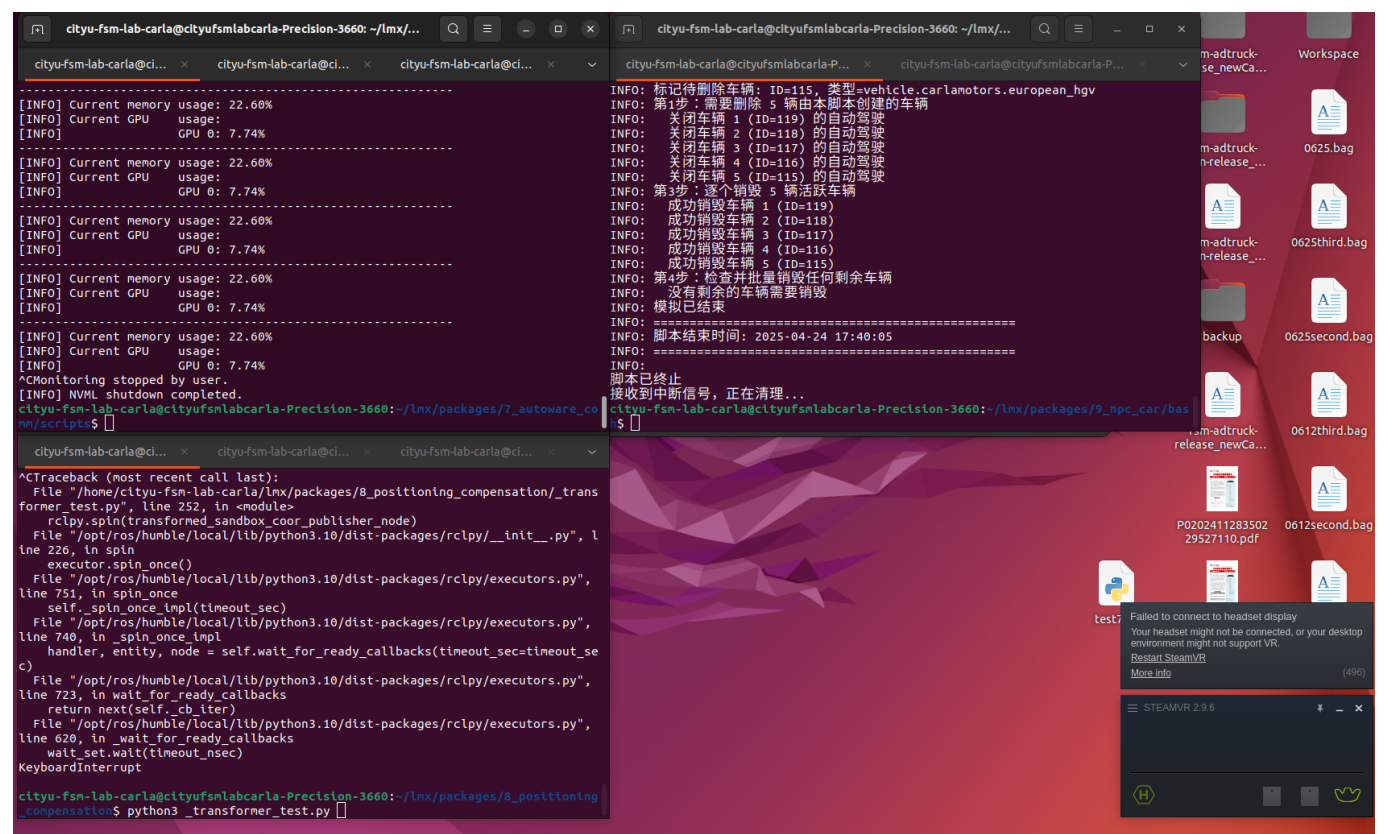


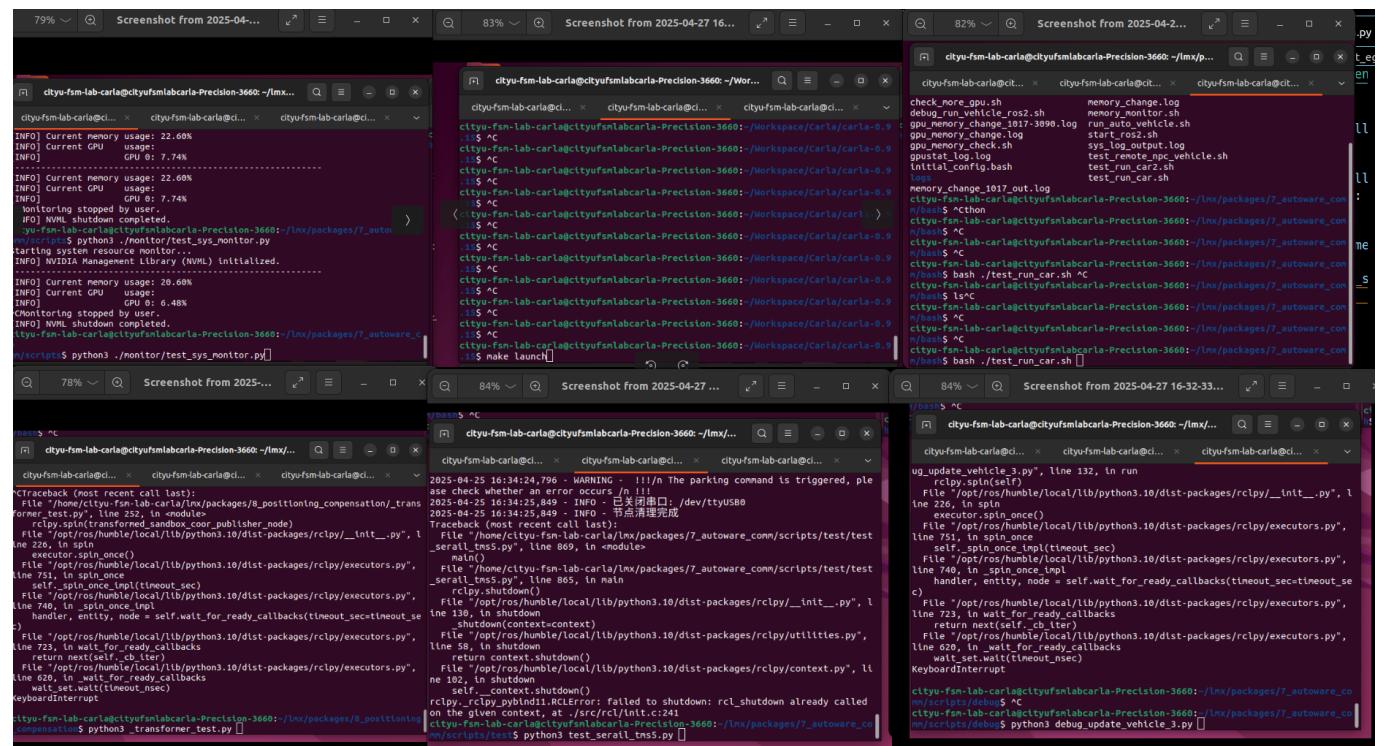
这是一个简化启动的操作手册

0.概述

在监控屏幕,可以看到以下几个terminal



以下是六个主要terminal · 每个terminal均处于按下enter即可启动的状态

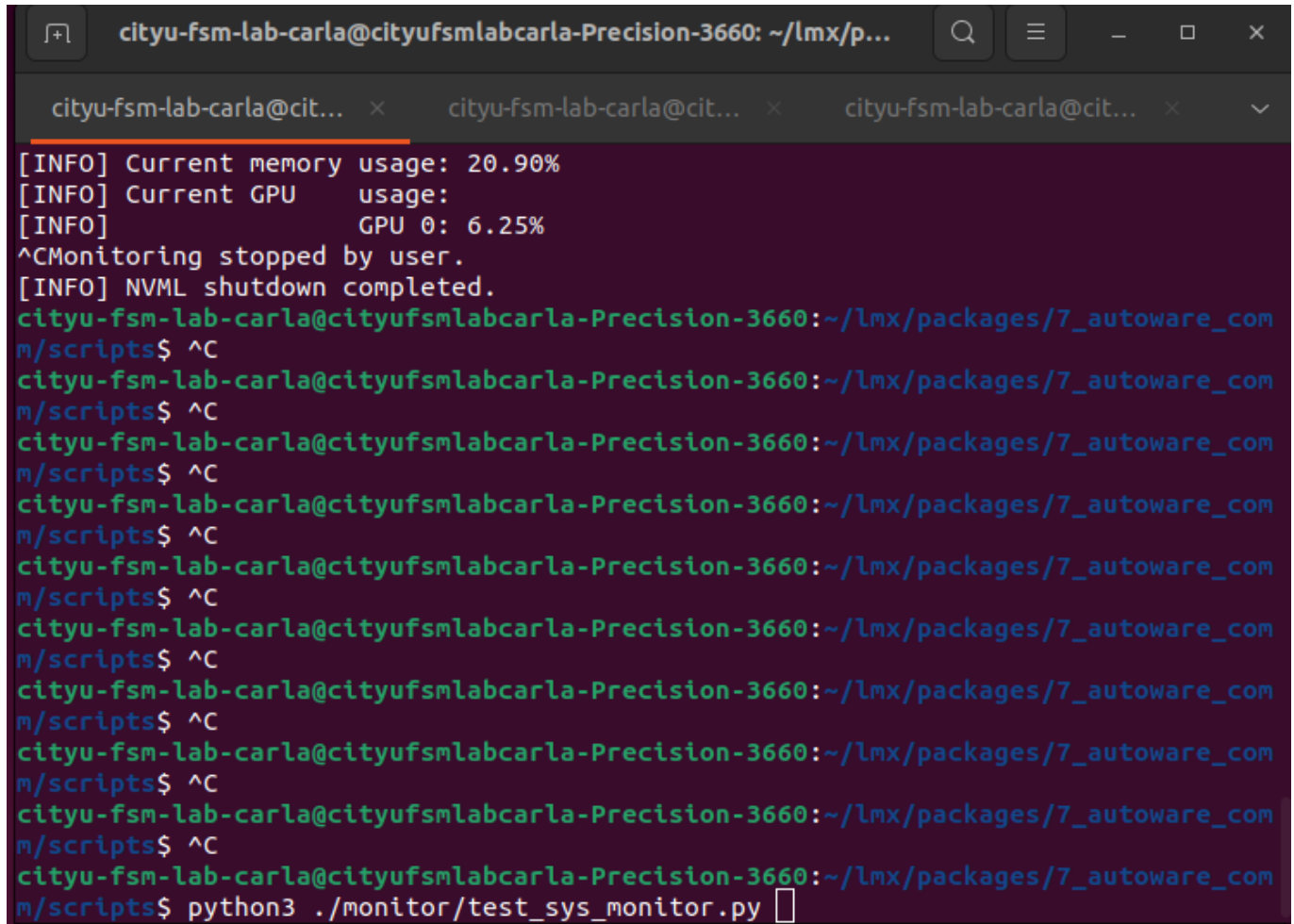


正常启动后的状态如下：

以下是系统启动顺序:

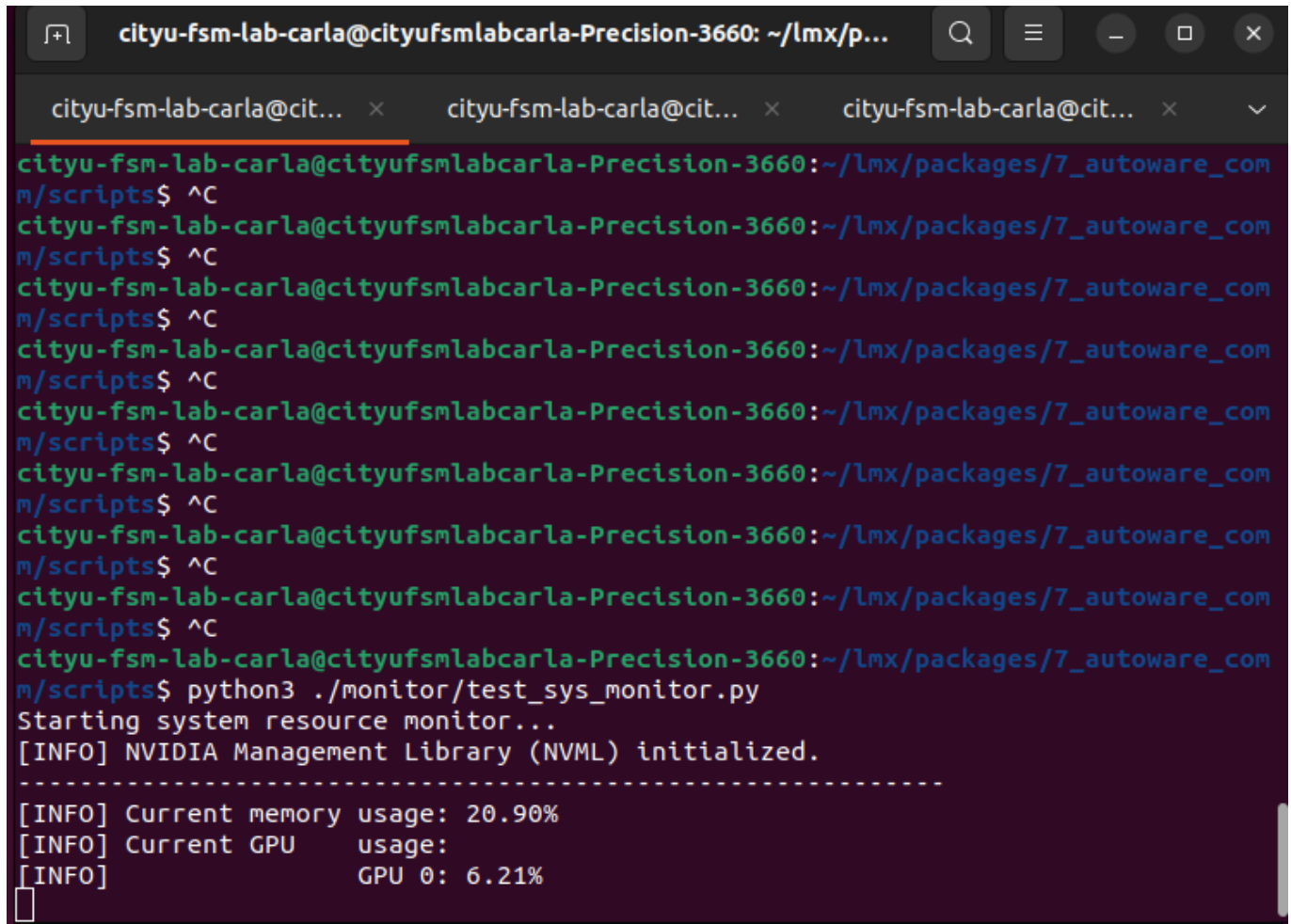
1. 启动系统资源监管器

在terminal的第一个tab，按下回车



```
cityu-fsm-lab-carla@cityufsm-lab-carla-Precision-3660: ~/lmx/p...
cityu-fsm-lab-carla@cit... x cityu-fsm-lab-carla@cit... x cityu-fsm-lab-carla@cit... x
[INFO] Current memory usage: 20.90%
[INFO] Current GPU usage:
[INFO] GPU 0: 6.25%
^C Monitoring stopped by user.
[INFO] NVML shutdown completed.
cityu-fsm-lab-carla@cityufsm-lab-carla-Precision-3660: ~/lmx/packages/7_autoware_com
m/scripts$ ^C
cityu-fsm-lab-carla@cityufsm-lab-carla-Precision-3660: ~/lmx/packages/7_autoware_com
m/scripts$ ^C
cityu-fsm-lab-carla@cityufsm-lab-carla-Precision-3660: ~/lmx/packages/7_autoware_com
m/scripts$ ^C
cityu-fsm-lab-carla@cityufsm-lab-carla-Precision-3660: ~/lmx/packages/7_autoware_com
m/scripts$ ^C
cityu-fsm-lab-carla@cityufsm-lab-carla-Precision-3660: ~/lmx/packages/7_autoware_com
m/scripts$ ^C
cityu-fsm-lab-carla@cityufsm-lab-carla-Precision-3660: ~/lmx/packages/7_autoware_com
m/scripts$ ^C
cityu-fsm-lab-carla@cityufsm-lab-carla-Precision-3660: ~/lmx/packages/7_autoware_com
m/scripts$ ^C
cityu-fsm-lab-carla@cityufsm-lab-carla-Precision-3660: ~/lmx/packages/7_autoware_com
m/scripts$ ^C
cityu-fsm-lab-carla@cityufsm-lab-carla-Precision-3660: ~/lmx/packages/7_autoware_com
m/scripts$ ^C
cityu-fsm-lab-carla@cityufsm-lab-carla-Precision-3660: ~/lmx/packages/7_autoware_com
m/scripts$ python3 ./monitor/test_sys_monitor.py
```

启动后状态如下:

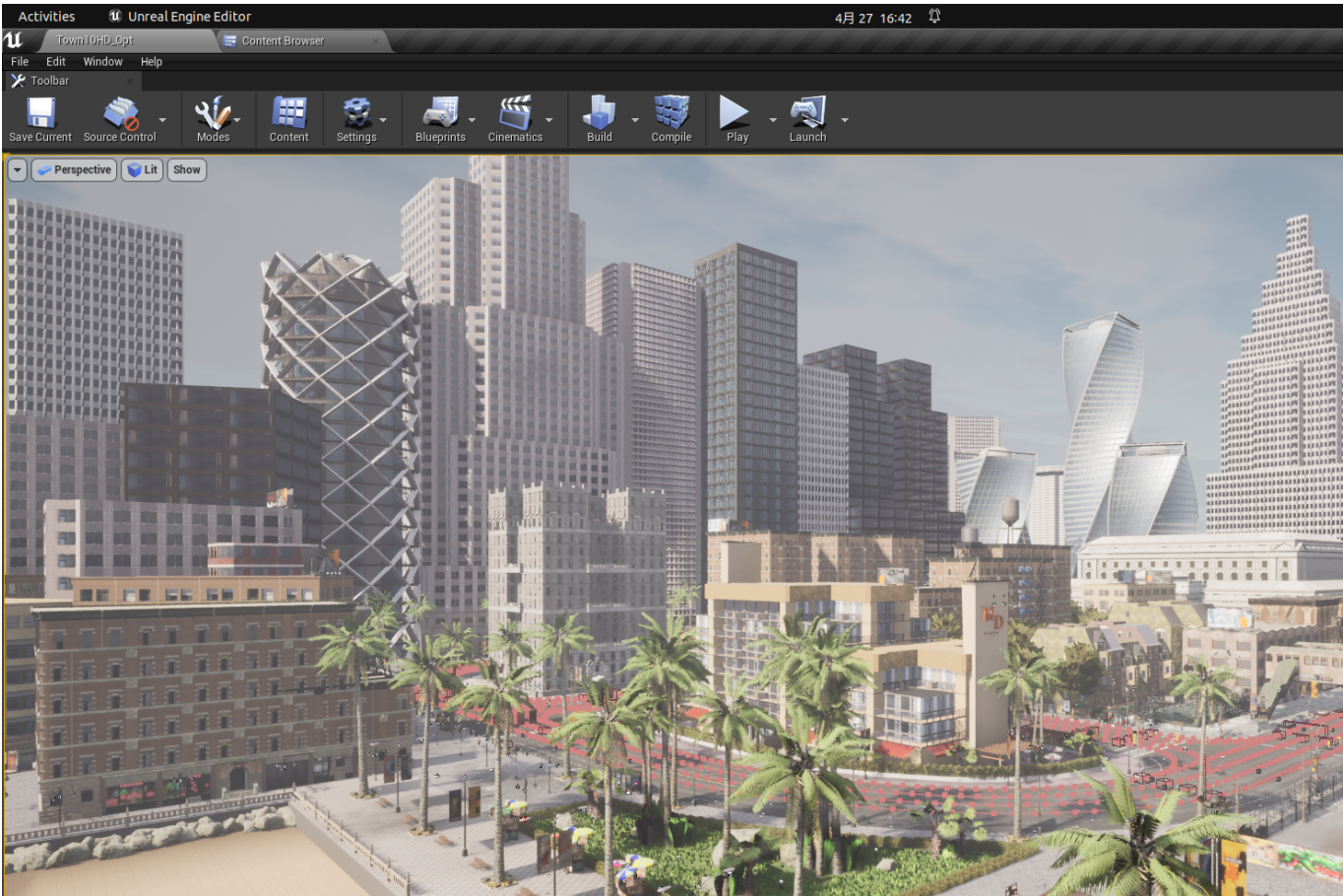
A terminal window titled 'cityu-fsm-lab-carla@cityufsm-lab-carla-Precision-3660: ~/lmx/p...' with three tabs. The terminal shows a series of '^C' commands followed by the execution of 'python3 ./monitor/test_sys_monitor.py'. The output indicates the system resource monitor is starting, NVML is initialized, and displays current memory usage at 20.90%, GPU usage at 6.21%, and GPU 0 usage at 6.21%.

```
cityu-fsm-lab-carla@cityufsm-lab-carla-Precision-3660:~/lmx/packages/7_autoware_com
m/scripts$ ^C
cityu-fsm-lab-carla@cityufsm-lab-carla-Precision-3660:~/lmx/packages/7_autoware_com
m/scripts$ ^C
cityu-fsm-lab-carla@cityufsm-lab-carla-Precision-3660:~/lmx/packages/7_autoware_com
m/scripts$ ^C
cityu-fsm-lab-carla@cityufsm-lab-carla-Precision-3660:~/lmx/packages/7_autoware_com
m/scripts$ ^C
cityu-fsm-lab-carla@cityufsm-lab-carla-Precision-3660:~/lmx/packages/7_autoware_com
m/scripts$ ^C
cityu-fsm-lab-carla@cityufsm-lab-carla-Precision-3660:~/lmx/packages/7_autoware_com
m/scripts$ ^C
cityu-fsm-lab-carla@cityufsm-lab-carla-Precision-3660:~/lmx/packages/7_autoware_com
m/scripts$ ^C
cityu-fsm-lab-carla@cityufsm-lab-carla-Precision-3660:~/lmx/packages/7_autoware_com
m/scripts$ ^C
cityu-fsm-lab-carla@cityufsm-lab-carla-Precision-3660:~/lmx/packages/7_autoware_com
m/scripts$ python3 ./monitor/test_sys_monitor.py
Starting system resource monitor...
[INFO] NVIDIA Management Library (NVML) initialized.
-----
[INFO] Current memory usage: 20.90%
[INFO] Current GPU      usage:
[INFO] GPU 0: 6.21%
```

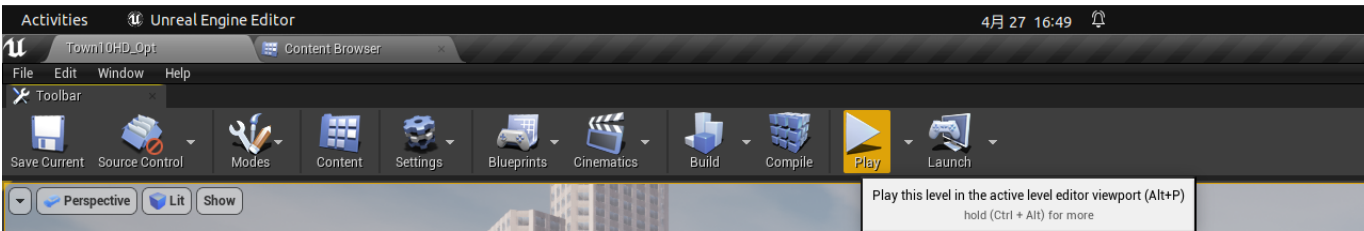
资源监管器可以在carla发生内存泄漏时将其自动强行杀死

2. 启动carla并激活模拟器

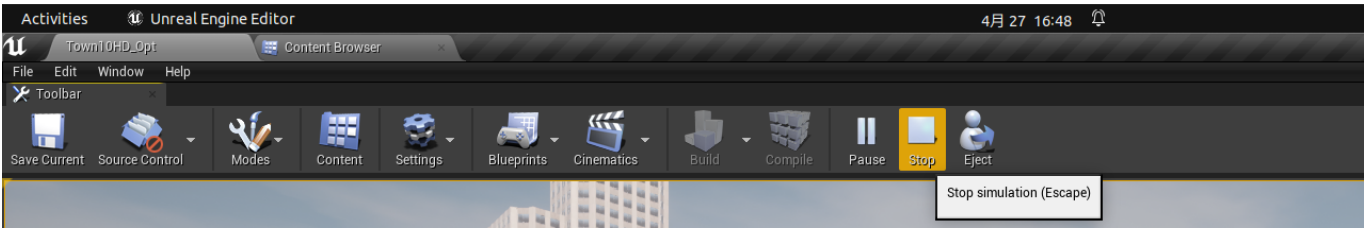
A screenshot of a terminal window titled "cityu-fsm-lab-carla@cityufsmmlabcarla-Precision-3660: ~/Works...". The terminal shows a series of 17 Ctrl+C (^C) commands entered at the prompt "cityu-fsm-lab-carla@cityufsmmlabcarla-Precision-3660:~/Workspace/Carla/carla-0.9.1". After the final Ctrl+C, the command "make launch" has been entered, and a cursor is visible at the end of the line. The terminal background is dark gray, and the text is light blue/green.



在carla正常启动后，按下屏幕上的三角paly键，等待激活服务器;



第一次激活需要很久，激活成功后的这次不可以使用（受限与carla的线程机制），需要关闭后二次激活。

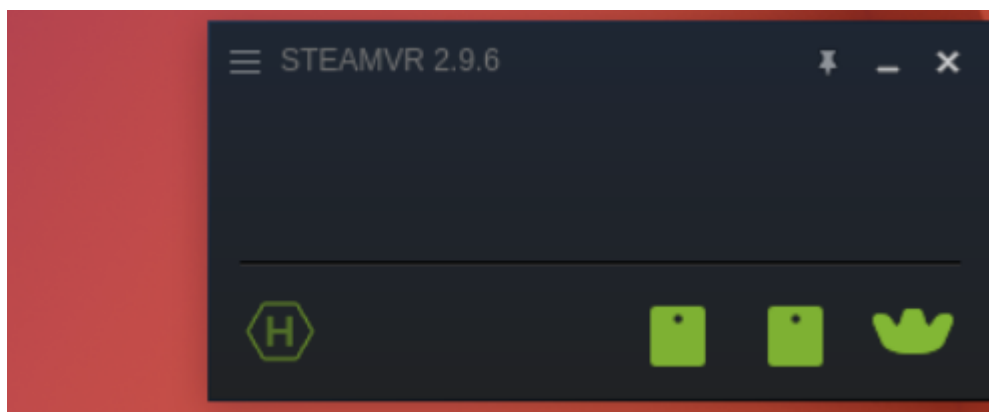


完成第二次激活后，在菜单栏处拖动carla将其移动到右侧屏幕（这样可以避免最小化，导致ubuntu将其进程级别降级）

3. 启动模拟

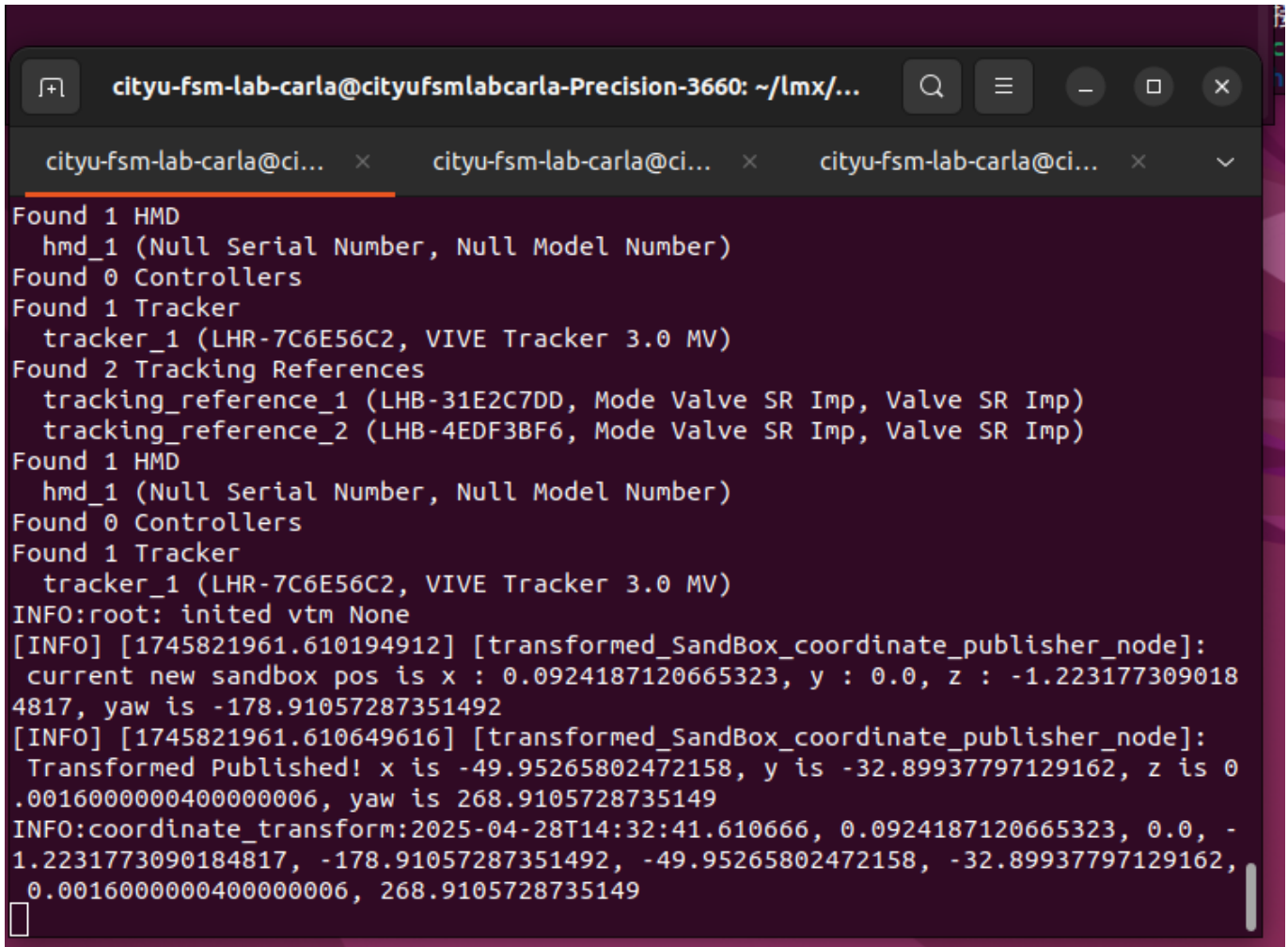
3.1 打开定位器

长按tracker中心直至其变绿,此时在屏幕右下角的steamr处应显示三个图表为绿色（可能会二绿一灰，此时移动一下tracker即可）



3.2 启动位姿转发

这是启动自动驾驶的关键,在左下这个terminal的第一个tab；按下回车，看到出现一条坐标消息，说明正常启动



```

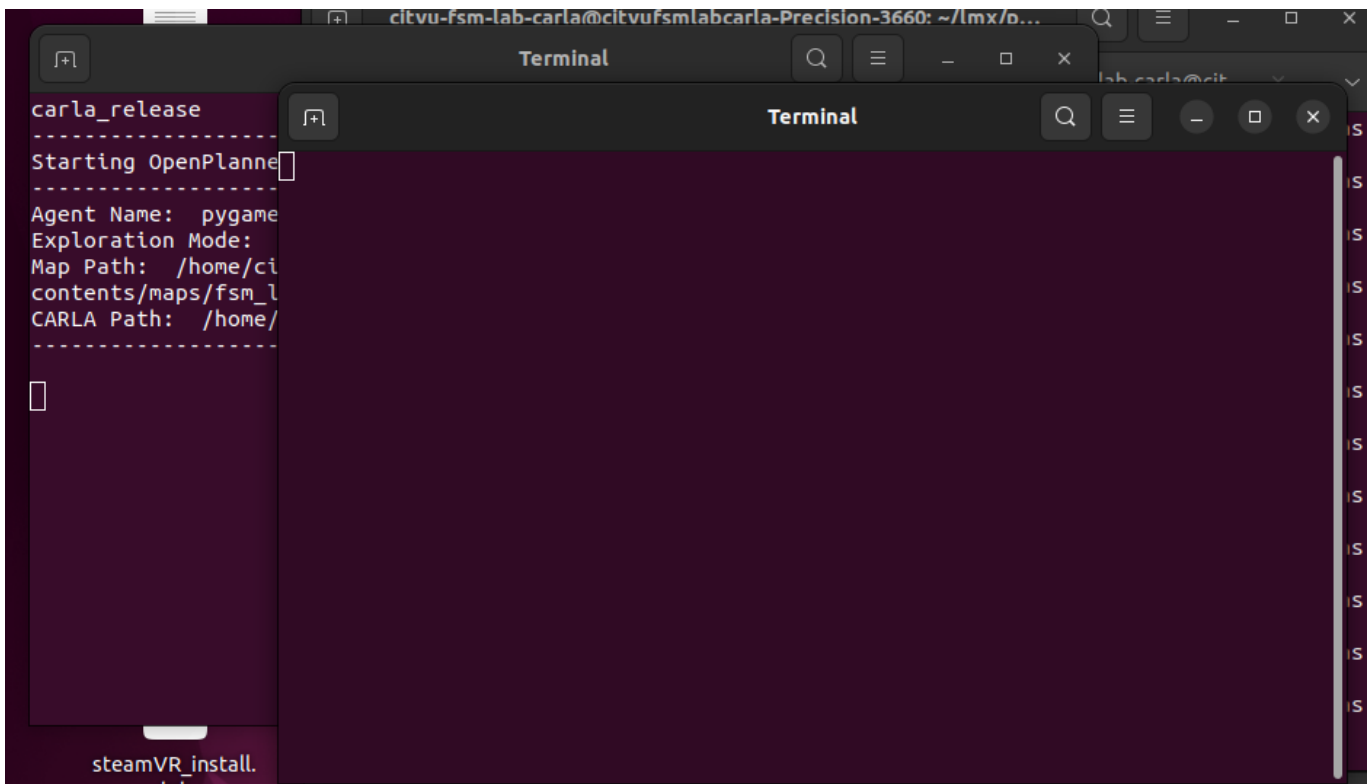
cityu-fsm-lab-carla@cityufsm-lab-carla-Precision-3660: ~/lmx/...
Found 1 HMD
  hmd_1 (Null Serial Number, Null Model Number)
Found 0 Controllers
Found 1 Tracker
  tracker_1 (LHR-7C6E56C2, VIVE Tracker 3.0 MV)
Found 2 Tracking References
  tracking_reference_1 (LHB-31E2C7DD, Mode Valve SR Imp, Valve SR Imp)
  tracking_reference_2 (LHB-4EDF3BF6, Mode Valve SR Imp, Valve SR Imp)
Found 1 HMD
  hmd_1 (Null Serial Number, Null Model Number)
Found 0 Controllers
Found 1 Tracker
  tracker_1 (LHR-7C6E56C2, VIVE Tracker 3.0 MV)
INFO:root: initd vtm None
[INFO] [1745821961.610194912] [transformed_SandBox_coordinate_publisher_node]:
  current new sandbox pos is x : 0.0924187120665323, y : 0.0, z : -1.223177309018
4817, yaw is -178.91057287351492
[INFO] [1745821961.610649616] [transformed_SandBox_coordinate_publisher_node]:
  Transformed Published! x is -49.95265802472158, y is -32.89937797129162, z is 0
.0016000000400000006, yaw is 268.9105728735149
INFO:coordinate_transform:2025-04-28T14:32:41.610666, 0.0924187120665323, 0.0, -
1.2231773090184817, -178.91057287351492, -49.95265802472158, -32.89937797129162,
0.0016000000400000006, 268.9105728735149

```

3.3 一键启动autoware与监控画面

在左上这个terminal的第三个tab，按下回车，会自动启动autoware，rviz与pygame;

会自动启动如下两个terminal，以启动autoware



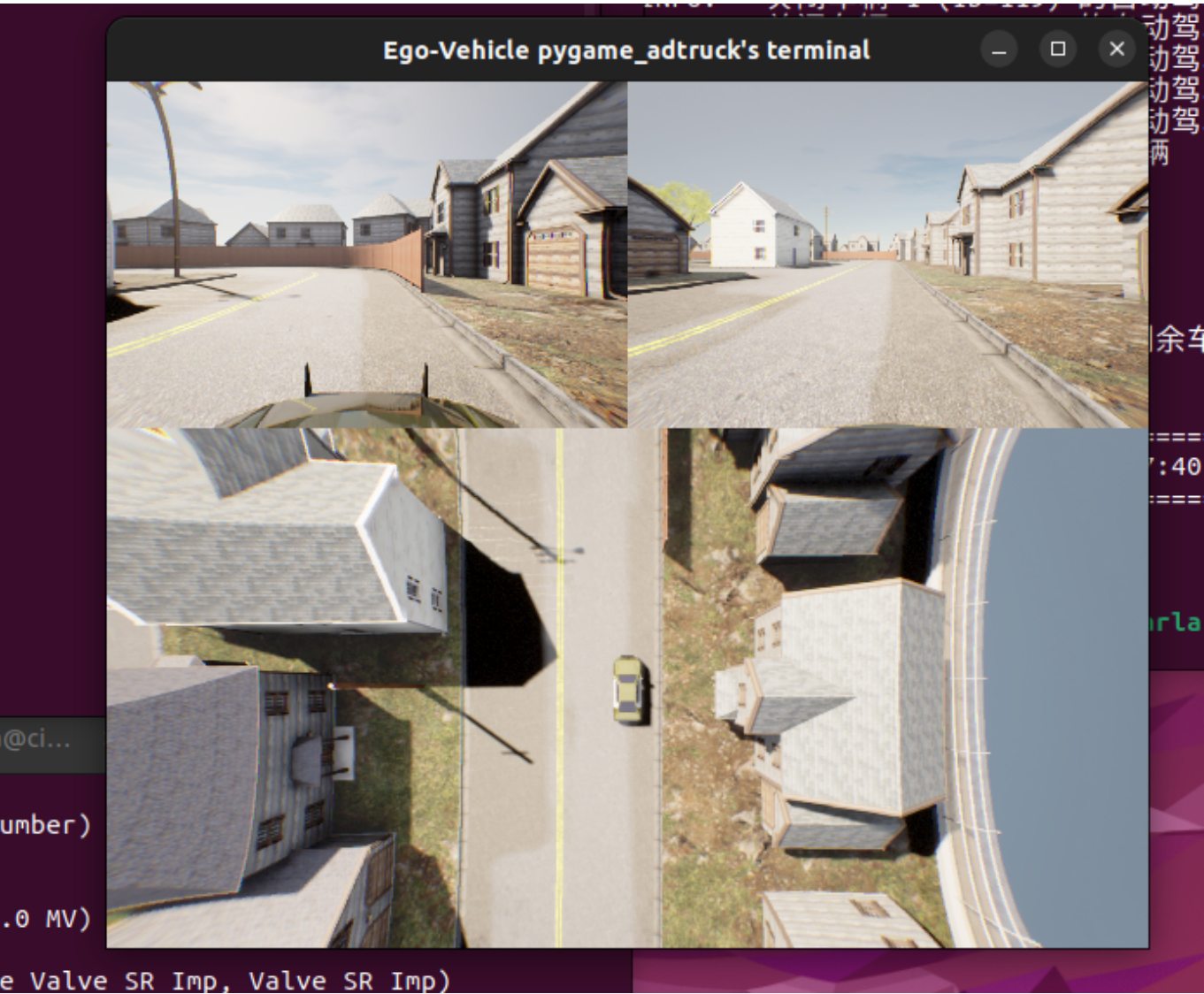
```

Terminal
carla_release
-----
Starting OpenPlanter
-----
Agent Name:  pygame
Exploration Mode:
Map Path:  /home/ci
contents/maps/fsm_l
CARLA Path:  /home/

Terminal

```

出现pygame窗口，说明小车在carla中初始化成功

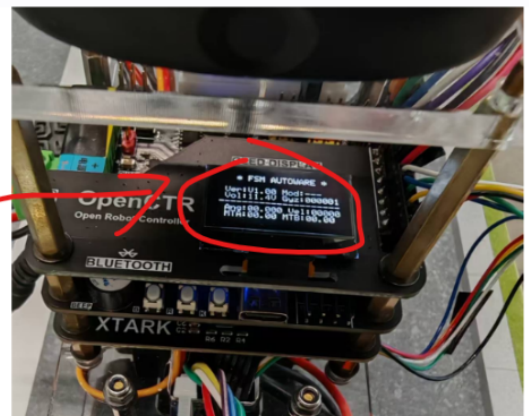
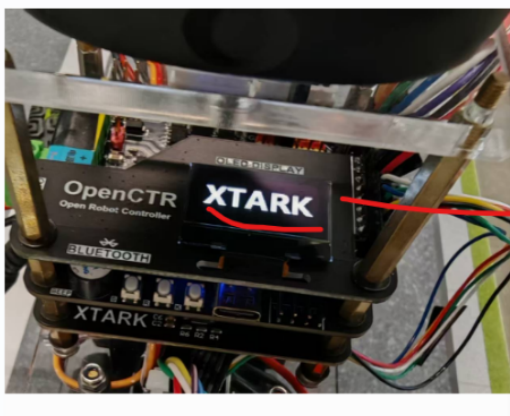


此过程需要等待较长的时间，等待出现如下图所示的状态（出现闪烁的车辆以及地图中心的坐标系中心图标）则说明已经成功启动。

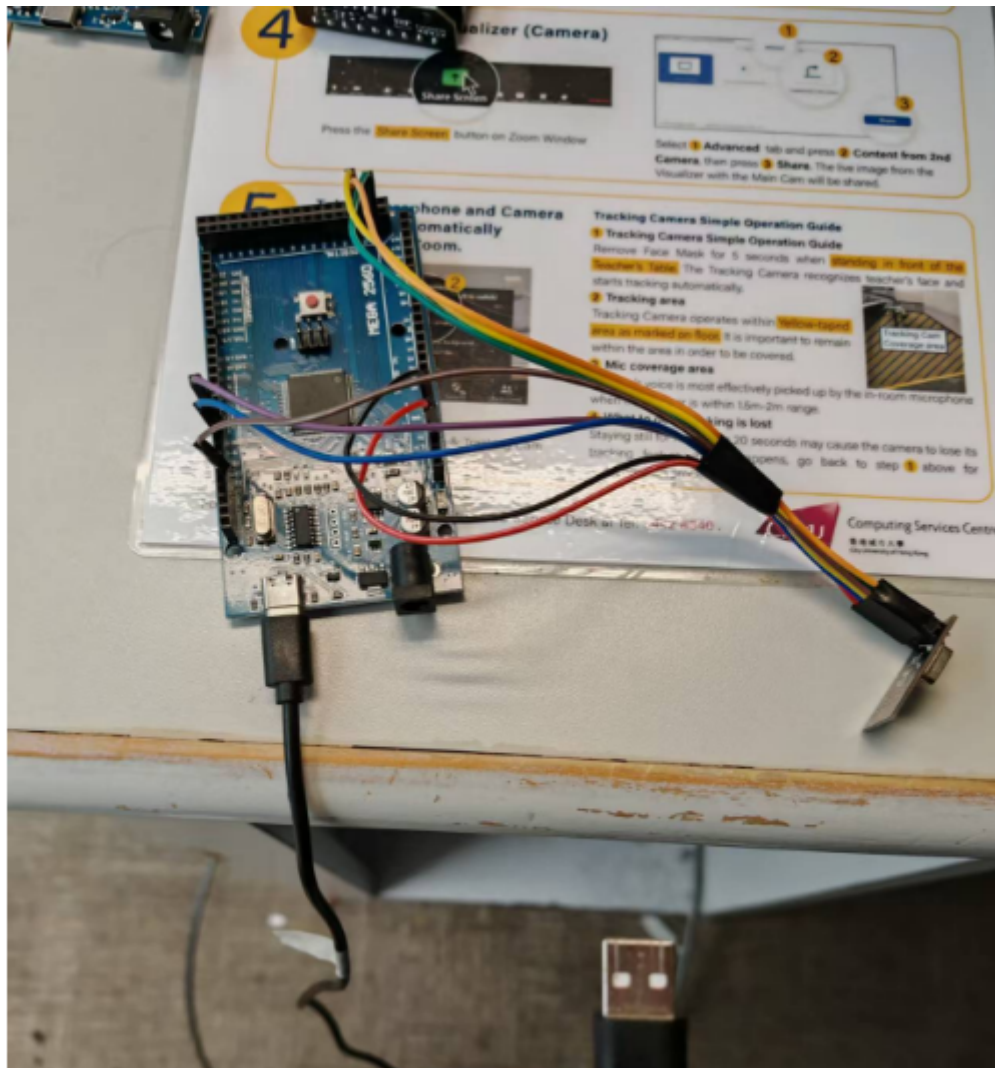


3.4 启动rc车

1. 启动rc小车 (car上的显示器进入到 fsm autoware界面)



2. 连接cmd转发设备：将usb端口连接到com4口（第一个usb口）





3. 启动转发脚本(左下terminal的第二个tab)并等待成功连接 (rc车会发出di~~ 一声长蜂鸣)

```

cityu-fsm-lab-carla@cityufsm-lab-carla-Precision-3660: ~/lmx/...
cityu-fsm-lab-carla@ci... x cityu-fsm-lab-carla@ci... x cityu-fsm-lab-carla@ci... x v
on the given context, at ./src/rc1/init.c:241
cityu-fsm-lab-carla@cityufsm-lab-carla-Precision-3660:~/lmx/packages/7_awareware_co
mm/scripts/test$ python3 test_serail_tms5.py
2025-04-29 14:53:02,574 - INFO - === 集成控制发布者启动于 20250429_145302 ===
2025-04-29 14:53:02,574 - INFO - 通信详情日志文件: logs/comm_details_20250429_14
5302.log
2025-04-29 14:53:02,574 - INFO - 时间戳CSV日志文件: logs/timing_log_20250429_145
302.csv
2025-04-29 14:53:02,575 - INFO - 参数配置: 端口=/dev/ttyUSB0, 波特率=115200, 超
时=0.5秒
2025-04-29 14:53:02,575 - INFO - 同步配置: 最大尝试次数=3, 超时=0.05秒
2025-04-29 14:53:02,575 - INFO - 成功打开串口: /dev/ttyUSB0
2025-04-29 14:53:04,577 - INFO - 当前模式: AUTO
2025-04-29 14:53:04,580 - INFO - 发送初始停止命令
2025-04-29 14:53:04,580 - INFO - 发送时间同步请求: 2 ms (相对时间戳), 尝试次数:
1/3
2025-04-29 14:53:04,581 - INFO - 发送到串口 - 时间戳: 3 ms, 速度: 0.0 m/s, 转向
角度: 0.0 度
2025-04-29 14:53:04,582 - WARNING - !!!/n The parking command is triggered, ple
ase check whether an error occurs /n !!!

===== Integrated Control Publisher =====
当前模式: AUTO
控制指令:

```

```

cityu-fsm-lab-carla@cityufsm-lab-carla-Precision-3660: ~/lmx/...
cityu-fsm-lab-carla@ci... x cityu-fsm-lab-carla@ci... x cityu-fsm-lab-carla@ci... x v
2025-04-29 14:53:04,582 - WARNING - !!!/n The parking command is triggered, ple
ase check whether an error occurs /n !!!

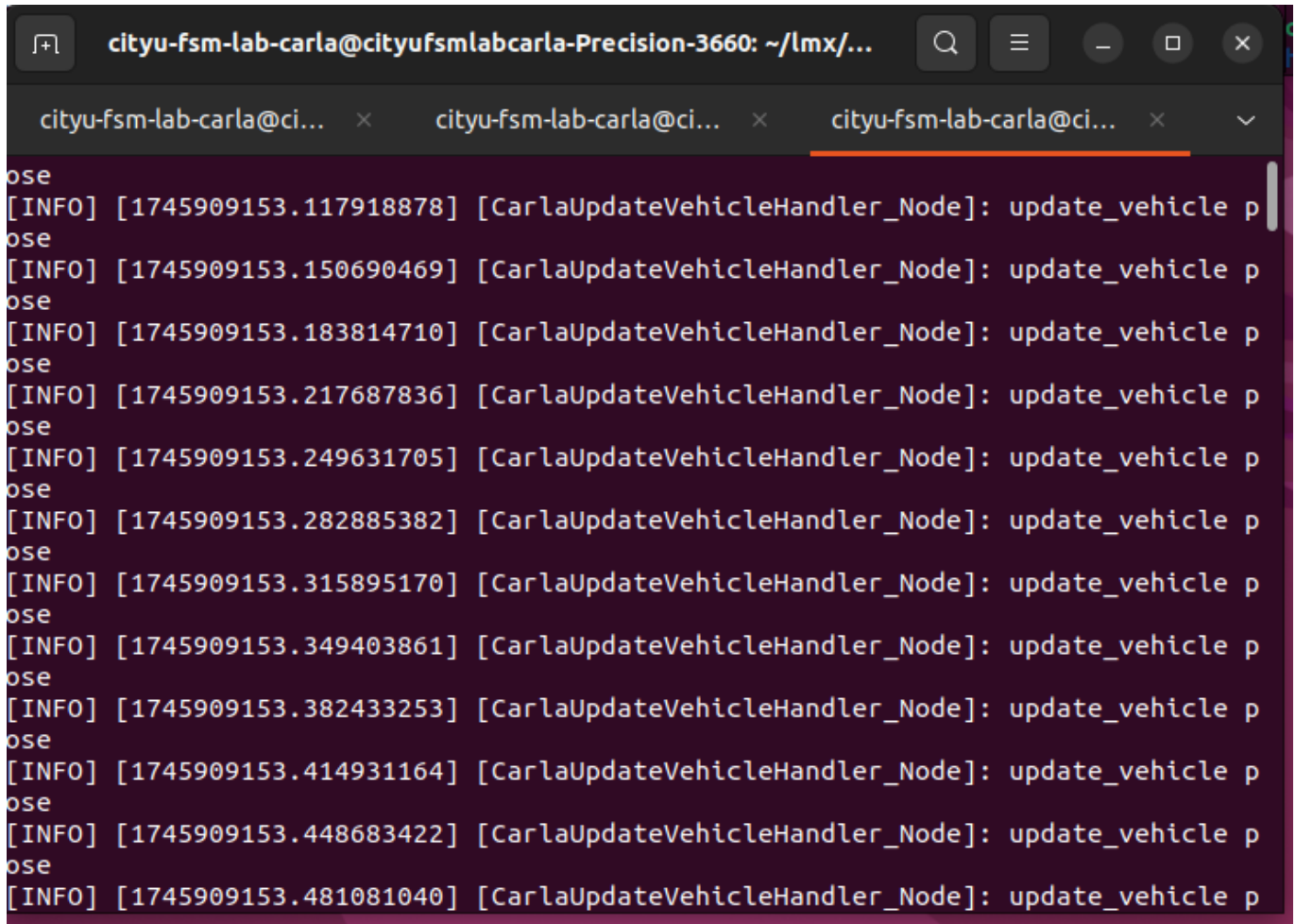
===== Integrated Control Publisher =====
当前模式: AUTO
控制指令:
- 按 'm' 切换控制模式 (AUTO <-> MANUAL)
- 手动模式下使用以下键盘控制:
  * 'w' 增加速度
  * 's' 减少速度
  * 'a' 向左转
  * 'd' 向右转
  * 'Space' 设置速度=0且转向角=0
- 按 'F5' 键手动触发时间同步
- 按 'Esc' 键退出程序
时间戳日志将记录到: /home/cityu-fsm-lab-carla/lmx/packages/7_awareware_co
mm/scripts/test/logs/timing_log_20250429_145302.csv
* 注意: 如果时间同步失败超过最大尝试次数, 将自动切换到手动模式并停车

=====
2025-04-29 14:53:04,588 - INFO - 时间同步成功 - PC时间: 2 ms, Arduino时间: 1091
ms, 偏移量: -1089 ms, 延迟: 9 ms
2025-04-29 14:53:04,609 - INFO - 命令接收确认 - PC时间戳: 3 ms, Arduino时间: 111
2 ms, 延迟: 29 ms

```

3.5 启动帧刷新

左下terminal的第三个tab



The screenshot shows a terminal window with three tabs. The active tab is the third one, titled 'cityu-fsm-lab-carla@cityufsm-lab-carla-Precision-3660: ~/lmx/...'. The terminal displays a series of log messages, each starting with '[INFO]' followed by a timestamp and the message '[CarlaUpdateVehicleHandler_Node]: update_vehicle p'. The timestamps are: 1745909153.117918878, 1745909153.150690469, 1745909153.183814710, 1745909153.217687836, 1745909153.249631705, 1745909153.282885382, 1745909153.315895170, 1745909153.349403861, 1745909153.382433253, 1745909153.414931164, 1745909153.448683422, and 1745909153.481081040. The messages are repeated 12 times in total.

完成上述三步，即可成功启动系统

4. 一件启动npc仿真

在右上第一个tab按下回车，启动自动臂章测试脚本

```
cityu-fsm-lab-carla@cityufsm-lab-carla-Precision-3660: ~/lmx/...
cityu-fsm-lab-carla@cityufsm-lab-carla-P... x cityu-fsm-lab-carla@cityufsm-lab-carla-P... x
结束时间: 2025年 04月 29日 星期二 14:48:24 CST
最近日志内容:
2025-04-29 14:48:19,097 - INFO - Turning off autopilot for vehicle 1 (ID=243)
2025-04-29 14:48:23,660 - INFO - Step 3: Destroying 1 active vehicles individually
2025-04-29 14:48:23,668 - INFO - Successfully destroyed vehicle 1 (ID=243)
2025-04-29 14:48:23,719 - INFO - Step 4: Checking and batch destroying any remaining vehicles
2025-04-29 14:48:23,719 - INFO - No remaining vehicles to destroy
2025-04-29 14:48:24,220 - INFO - Destroyed 1 vehicles
2025-04-29 14:48:24,220 - INFO - Cleaning up ROS2 resources...
2025-04-29 14:48:24,460 - INFO - ROS2 resources cleaned up
2025-04-29 14:48:24,460 - INFO - Cleanup completed
2025-04-29 14:48:24,460 - INFO - Simulation ended
2025-04-29 14:48:24,460 - INFO - =====
===
2025-04-29 14:48:24,460 - INFO - Script end time: 2025-04-29 14:48:24
2025-04-29 14:48:24,460 - INFO - =====
===
2025-04-29 14:48:24,460 - INFO - Script terminated
===== 启动器结束 =====
cityu-fsm-lab-carla@cityufsm-lab-carla-Precision-3660: ~/lmx/packages/9_npc_car/bas
h$ bash integrated_scenario_launcher.sh
```

```
cityu-fsm-lab-carla@cityufsm-lab-carla-Precision-3660: ~/lmx/...
cityu-fsm-lab-carla@cityufsm-lab-carla-P... x cityu-fsm-lab-carla@cityufsm-lab-carla-P... x
h$ bash integrated_scenario_launcher.sh
===== 检查环境 =====
已找到: Python 3.10.12
===== 集成场景测试启动器 =====
启动时间: 2025年 04月 29日 星期二 14:56:21 CST
日志文件: logs/scenario_run_20250429_145621.log
基本参数:
主机: 127.0.0.1
端口: 2000
TM端口: 8100
NPC车辆数量: 5
随机种子: 42
时间和距离参数:
发送engage前等待时间: 2.0 秒
在路点处等待时间: 5.0 秒
位置检查间隔: 0.5 秒
最大位置等待时间: 30.0 秒
目标检测距离阈值: 8.0 米
起点检测距离阈值: 10.0 米
键盘控制:
在NPC测试期间按下 'T': 触发碰撞恢复模式 (移除NPC车辆并恢复导航)
在圆形循环期间按下 'R': 重新启动NPC测试
按回车键继续...

```

在此脚本中，按下回车即可启动npc_test；当结束npc_test结束后，会开始circle绕外圈循环

